Highlights

Draft paper base bach thesis

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ABSTRACT

Here goes the abstract
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1. Introduction and literature review

Public transport offer analysis can result a very complicated task for governmental entities. If the analysis is performed over real data recovered from trips across days, multiple month data window with same configuration is required in order to obtain statistically accurate results. The binding constraint in this case, however, is not the volume of the data stored, but rather the dependency a small change in offer configuration can shift the data results towards inconsistencies due to that derivative. For this specific problem. multiple entities relies on simulation techniques for the purpose of analyzing without real deployement. Cases like [1–3] present different simulation techniques applied in the public transport domain.

This work presents two modelling solutions based on the MAS paradigm. Using the multi agent paradigm in the transport simulation field is a well known and implemented technique that has increase among the last years [4].

Some of them take advantage of the cognitive contribution that the use of certain paradigms, like BDI (Belief Desire Intention) brings.

Nontheless, the transport simulation problems are indeed a type of problem with a strong variable and dynamic environment, so the use of the ABM is appropiate regardless of the agent taxonomy itself. This disctintion is given depending on the scope the simulation is aiming and the set of data avaliable.

At a microscopic level, the BDI can represent individual behaviour

In a mesoscopic scope, that

(Aqui el posicionamiento del trabajo dentro del SOTA y su "aportacion" (el modelado del sistema))

Tendré que hablar de que el enfoque de la simulacion es una mezcla entre mesoscopico y microscopico. Al fin y al cabo es demostrar que para el objetivo de análisis a nivel vehicular de un autobus, ambos son enfoques validos mediante el modelado que excluye el nivel cognitivo de la entidad pasajero, convirtiendolo meramente en un valor agregado a la carga del autobus, pero manteniendo la fidelidad estadística con los datos observables por la EMT.

This work focus its scope over the bus capacity of demand absortion, so the mesoscopic approach seems to suit better, specially considering the computational cost it inquires the indiscrecionary granularity augment. However, a mixed scope can be considered in one of the two proposed solutions, as it models passengers as agents. In this case, the context will still be mesoscopic, due to the fact that the agent itself is not being modelled with a specific BDI architecture, but its just been used as a statistical count for bus capacity.

P3 MAS dentro de la simulación de transporte. Aquí entra tu micro/meso, pero como marco heredado, no como aportación. Establece los tres niveles clásicos macroscópico agregado, mesoscópico por paquetes, microscópico por individuo y sitúa el MAS en el microscópico. Cita MATSim y BusMezzo como los representantes. Este párrafo existe para montar el siguiente. P4 La dependencia oculta del enfoque

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microscópico. El párrafo bisagra, y el que más te va a costar. El argumento: modelar al pasajero como agente exige darle un destino, y el destino no está en los datos APC. La literatura resuelve eso estimando matrices OD IPF con matriz semilla, formulaciones bayesianas y cada método arrastra sus supuestos. Aquí citas la literatura OD (IPF-null, Ji et al., el bayesiano de 2024). La frase clave: la granularidad del modelo está acoplada a un problema de estimación que el objetivo analítico no necesariamente requiere. P5 El hueco. Cuando el objetivo es el comportamiento del vehículo y no el itinerario del viajero, reconstruir la OD es resolver un problema más difícil que el planteado. Y no existe en la literatura un criterio, medible sobre datos, para decidir hasta dónde llega la agencia que puedes conceder. Parunak et al. (1998) dan la versión cualitativa; la multi-level ABM aborda la agregación como aproximación, no como equivalencia. P6 La aportación. Tres frases, una por pieza, y en este orden: El criterio: el nivel de agencia admisible está acotado por la granularidad de los datos de calibración; por encima de esa cota el comportamiento del agente es infalsable. La consecuencia de diseño: dos arquitecturas idénticas en distribución por construcción, una con pasajero reificado y otra con demanda agregada en la parada, que permiten aislar el coste de la reificación sin confundirlo con una aproximación. La validación: equivalencia distribucional contra datos reales de EMT por parada y franja, con caracterización explícita de dónde y por qué mecanismo falla el modelo. P7 Estructura del documento. Dos frase

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2. Proposed solutions: two different approaches

As for the purpose of this work, no cognitive reasoning is represented neither for the bus nor the passenger. Thus, the emergent intelligence comes from the interaction between agents and environment, as per Brooks [5], that emergent behaviour guarantees an agency right itself.

2.1. Agents and protocols

The agents defined in this scenarios are formalized as reactive agents. This decision
In the two proposed implementations, three agents are defined:

- bus
- stop

- Passenger

3. Data calibration

The system will rely on parametric data obtained through analysis of empirical 8 month operational window timespan.

The data covers all stop related information for every time in a day a bus reaches a stop.

In this work, as per cardinality constraints due to not reaching enough statistical data for covering all scenarios, only the labour days are included in the parametric analysis. Moreover, the freeday data is also excluded from the analysis, as it changes the behaviour of the buses occupation. The time series anal

3.1. Attributes to calibrate

The system is modelled for receiving four attributes.

For every stop, disaggregated into its timeslots, it expects distribution fitted parameters for both boarding and alighting processes.

The boarding value represents the number of people arriving to a stop each hour, and it will use it as a passengers generator ratio.

On the other hand, the alighting values are modelled as independent samples per bus arrival. Each time a bus reaches a stop, the system will pick a random sampling out of the distribution corresponding to the combination (stop-timeslot). This modelling decision is one of the reasons the use of OD matrixes is discarded in this work. As having APC data from the buses, the reconstruction of individual paths from boarding to alighting can be a more complex task than the core behaviour of the simulation itself **Creo que puede ser overclaim y no se si deberia ir en el SOTA esto.**. Also, when no heuristic to build the matrix is declared, IPF [**?**] method can be used with a NULL seeding [**?**]. As in this work no external surveys or data at all to construct OD matrixes are available, the use of empirical fitted distributions suits as a counter-option.

Things to mention: Why only labour, why no freeday, why no possibility of decouple stationality and analysis via temporal series

Also, mention the chosen distribution families, the why use of negbins and geoms, cause the variance > mean (poisson is variance == mean) Also, mention the NO OD data, and as i chose not to infer with IPF NULL seeding, its constructed by independent consideration of boarding and alighting.

3.2. Parametric implementation

4. Implementations and results

Both proposed architectures run in the dedicated agent platform GAMA [6]. The election is based on both the inclusion of FIPA inboxing, allowing implementation-defined conversations, and a headless batch mode that improves performance across runs.

4.1. Simulation parameters

For the simulation characterization, the layout of route Express 01 is used. The route operates between the centric square *Cibeles*, located in the city center of the Spanish's capital, and *La Peseta*, located in the southern neibourhood of Carabanchel. The route has 15 and 12 stops for each direction respectively, and its core behavour reflects the day by day of madrid's workers: Going to the center earlier in the morning to work and coming back home after lunch. The route's prime time in Cibeles's direction comes between 7 and 9 in the morning, boarding passengers in the first stops of the trip, and alighting them at the final ones. The same behaviour is inversed in La Peseta's direction, presenting its prime time after 3 pm, capturing the tendency of workers to return to its homeplace.

The specific parameterization used is shown in table 1.

Before the results are presented, some specifications about the simulations itself are introduced. The field `Extra dwell time per passenger` is used with both a premise and a comparison.

Table 1
Simulation parameters

Parameter	Value
Line	E01 (Cibeles-La Peseta)
Total physical buses	12 buses
Total trips	163 trips
Initial TimeSlot	5
Extra dwell time per passenger	5 seconds
Simulation time step	2 seconds
Minimum protocol dwell ($E_{\min}$)	4 * Simulation time step
Monte Carlo replicas	1000

First, it assumes that the boarding and alighting processes happen in parallel, as in regular bus operations those occur in different physical doors.

Secondly, the dwell extra time (E) will be the result of the following equation:

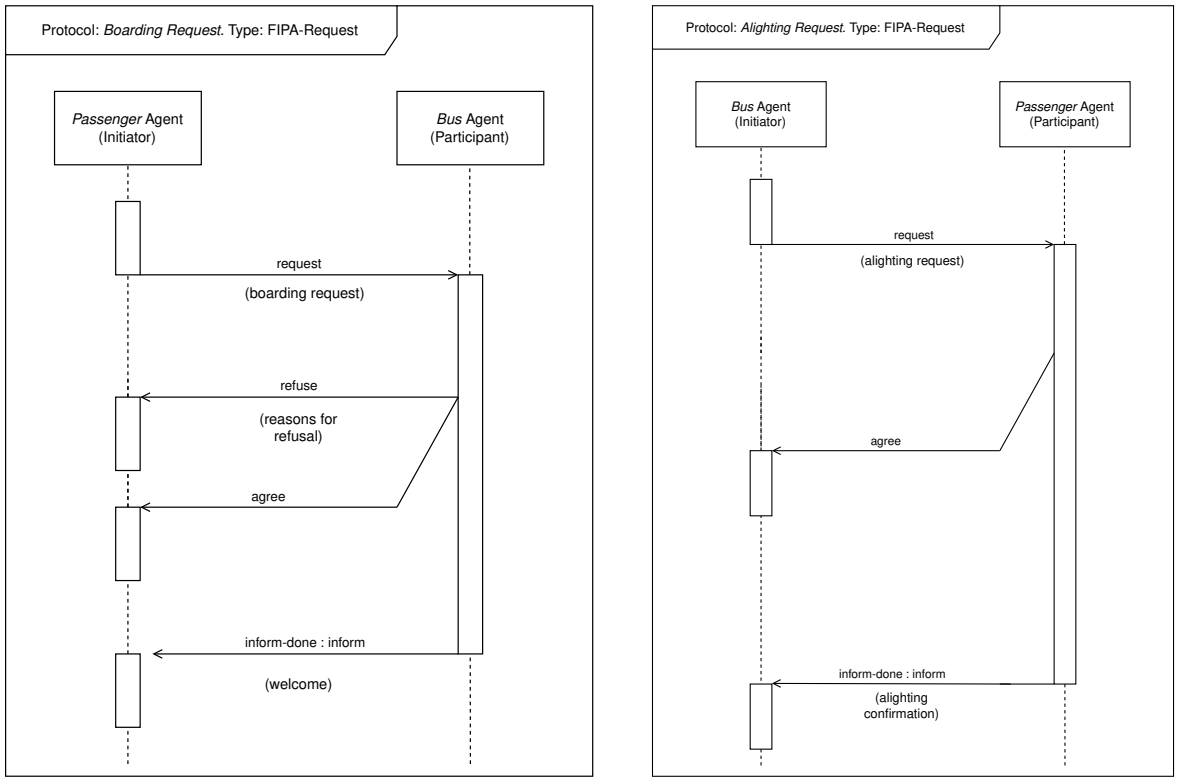
$$E = \max\big(\max(b,\, a)\, t_p,\; E_{\min}\big) \tag{1}$$

where b and a are the numbers of boarding and alighting passengers at the stop, t_p is the extra dwell time per passenger, and $E_{\min}$ is the minimum dwell imposed by the interaction protocol.

In relation to the necesary input files required to run the simulation,

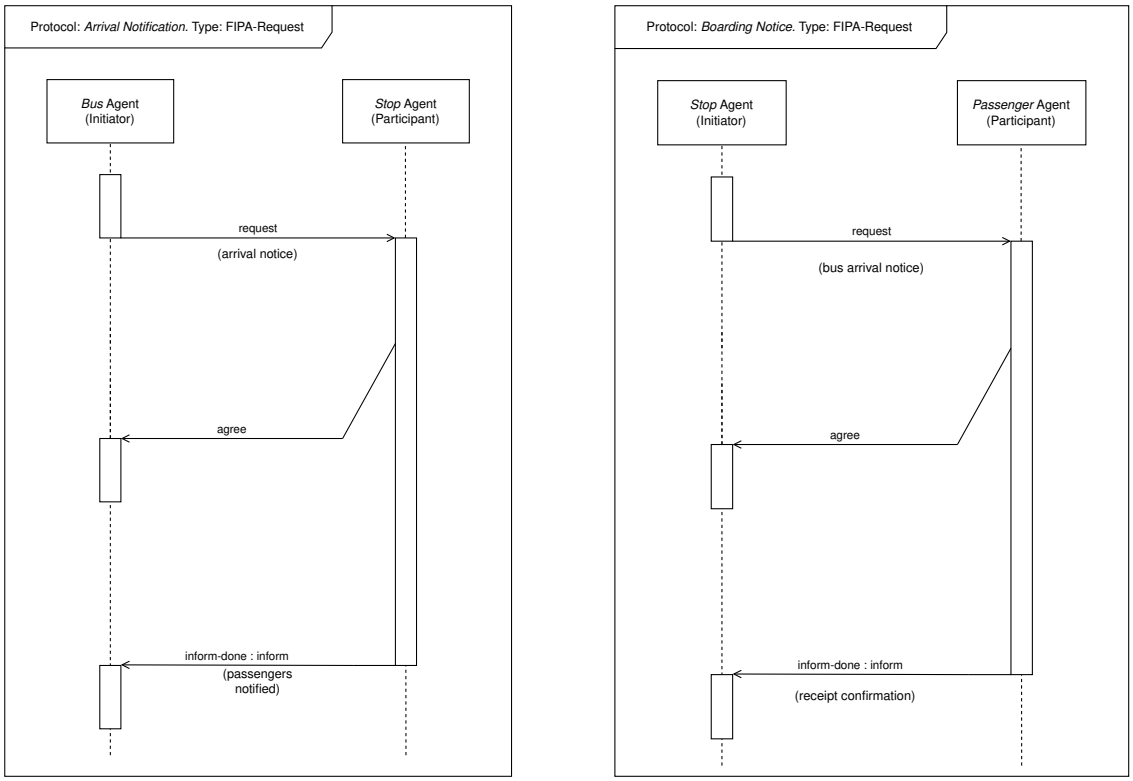
4.2. Results

5. Conclusions



(a) Boarding request (b) Alighting request

Figure 1: FIPA-Request interaction protocols for passenger requests.



(a) Boarding request (b) Alighting request

Figure 2: FIPA-Request interaction protocols for a bus arriving to a stop.

6.

CRediT authorship contribution statement

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